

## PRIMARY FINDING

**Diagnosis:** PITUITARY  
**Confidence:**  75.6%  
**Risk Level:**  MODERATE  
**Processing:** 8.33s (TTA 6-view ensemble)  
**Method:** EfficientNet-B3 + ResNet50+CBAM + DenseNet121

| Class      | Probability | Bar Chart              | Notes     |
|------------|-------------|------------------------|-----------|
| Glioma     | 6.9%        | <div><div></div></div> |           |
| Meningioma | 10.4%       | <div><div></div></div> |           |
| No Tumor   | 7.2%        | <div><div></div></div> |           |
| Pituitary  | 75.6%       | <div><div></div></div> | ← Primary |

| Model           | Confidence | Bar                    |
|-----------------|------------|------------------------|
| EfficientNet-B3 | 92.7%      | <div><div></div></div> |
| ResNet50+CBAM   | 40.1%      | <div><div></div></div> |
| DenseNet121     | 88.2%      | <div><div></div></div> |
| Ensemble (TTA)  | 75.6%      | ← FINAL                |

## CLINICAL INTERPRETATION

Findings raise the possibility of a pituitary region lesion, potentially a pituitary adenoma or sellar tumour. Dedicated pituitary protocol MRI, visual field assessment, and endocrinology referral are recommended. Hormonal panel (prolactin, GH, ACTH, TSH, FSH/LH) should be evaluated. Correlation with clinical symptoms (headache, visual changes, endocrine dysfunction) is essential.

## RECOMMENDATIONS

1. Dedicated pituitary protocol MRI (thin coronal slices)
2. Endocrinology referral for hormonal evaluation
3. Visual field examination (Goldman / Humphrey perimetry)
4. Ophthalmology review if bitemporal hemianopia suspected

■ **DISCLAIMER:** This report is generated by NeuroScan AI, an AI-assisted screening tool. It is intended for informational and research purposes only and does **NOT** constitute a medical diagnosis. All findings must be reviewed and confirmed by a qualified radiologist or neurologist before any clinical decision is made. Individual model performance may vary across patient populations and imaging protocols. Reported metrics: AUC 0.9996, Accuracy 98.78%, Sensitivity 98.68%, Specificity 99.59% (test set n=1,311 images). Report generated: 2026-04-20 15:55:19 IST | Report ID: RPT-7646B9D2